SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I (AJAY J / 192511016)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

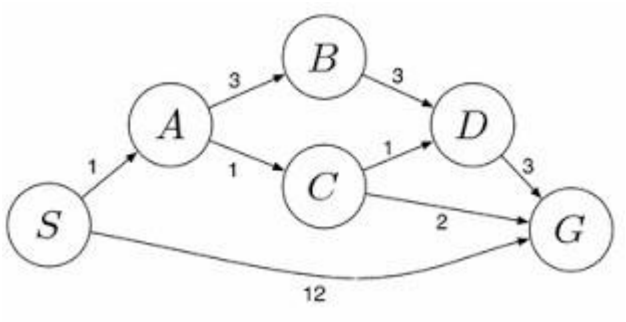
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

SOLUTIONS

1. Water Jug Problem

You are given a 4-gallon jug and a 3-gallon jug. The objective is to get exactly 2 gallons of water in the 4-gallon jug.
Solution:

Step	4-Gallon Jug	3-Gallon Jug	Action
Initial	0	0	Both jugs are empty
1	0	3	Fill the 3-gallon jug
2	3	0	Pour water into the 4-gallon jug
3	3	3	Fill the 3-gallon jug again
4	4	2	Pour water into the 4-gallon jug until it is full
5	0	2	Empty the 4-gallon jug
6	2	0	Pour the remaining 2 gallons into the 4-gallon jug

Result: The 4-gallon jug contains exactly 2 gallons.

2. Mars Rover Agent

i. Percepts
- Camera images
- Soil and rock samples
- Temperature readings
- Terrain information
- Obstacle detection
- Battery level

ii. Operating Environment
- Partially Observable
- Deterministic
- Sequential
- Dynamic
- Continuous
- Single Agent

iii. Actions
- Move forward and backward
- Turn left and right
- Collect soil and rock samples
- Drill rocks
- Capture images
- Transmit data to Earth

iv. Performance Measure
- Number of samples collected
- Accuracy of analysis
- Distance explored

- Energy efficiency
- Successful communication
- Safe navigation without collisions

v. Suitable Agent Architecture
Model-Based Goal-Based Agent
Reason:
A Mars Rover must remember previous observations, build a model of its surroundings, and plan actions to achieve scientific goals while operating in an unknown environment.

3. 8-Queens Problem

Problem Formulation
Initial State
An empty 8 × 8 chessboard.
State Space
All possible arrangements of eight queens on the chessboard.
Operators
Place one queen in a safe position in the next column.
Goal State
Place eight queens so that:

- No two queens are in the same row.
- No two queens are in the same column.
- No two queens are on the same diagonal.

Path Cost
Each queen placement has a cost of 1.
Total path cost = 8.
Search Strategy

- Backtracking Search
- Depth-First Search (DFS)

4. OLA Cab Booking Problem

Type of Problem-Solving Agent
Goal-Based Agent
Reason
The customer's goal is to travel from the source to the destination by selecting the most suitable cab based on comfort, availability, and cost.
Pseudocode
START

Input source location
Input destination

Display available cab types
(Mini, Micro, Sedan, Shared, Prime)

User selects a cab

Calculate fare

```
IF cab is available THEN
   Book the cab
   Display driver details
   Start trip
ELSE
   Display "Cab Not Available"
END IF

STOP
```

5. Uniform Cost Search (UCS)

Problem Statement

A logistics company wants to transport packages from warehouse S (Start) to warehouse G (Goal) at the minimum travel cost.

Uniform Cost Search Algorithm

1. Start from the source node S.
2. Insert S into the priority queue with cost 0.
3. Remove the node with the lowest cumulative cost.
4. If the removed node is the goal node G, stop.
5. Otherwise, expand its neighboring nodes.
6. Add each neighbor to the priority queue with its cumulative cost.
7. Repeat until the goal node is reached.

Pseudocode

```
UCS(Start, Goal)

Create Priority Queue

Insert(Start, Cost = 0)

WHILE Queue is not empty

   Remove node with lowest cost

   IF node = Goal THEN
      Return Path
   END IF

   FOR each neighboring node

      NewCost = CurrentCost + EdgeCost

      Insert neighbor with NewCost

   END FOR

END WHILE
```

Advantages

- Finds the least-cost path.
- Complete algorithm.
- Optimal when all edge costs are positive.

Disadvantages

- Requires more memory.
- Slower for very large graphs.